

The Entropic Cost of Global Consistency

Minimum-Entropy Couplings Under Exact Gluing Constraints

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Abstract

Minimum-entropy coupling asks for a joint distribution of smallest Shannon entropy among all couplings of prescribed marginals. In many local-to-global problems, however, the coordinates of a coupling are not semantically independent: local states may contain copies of the same overlap variable, and a valid global realization must identify those copies samplewise. We isolate the entropy cost of imposing that exact gluing condition.

For finite local state spaces X and Y equipped with restriction maps to a common overlap B , let κ be the ordinary minimum-entropy coupling of the local-state distributions and let χ_B be the minimum entropy among couplings satisfying equality of the two overlap copies almost surely. Their difference

$$\Gamma_B := \chi_B - \kappa$$

is the *gluing entropy premium*. When the two local distributions agree on the overlap, we prove the exact fiberwise formula

$$\chi_B = H(B) + \sum_b P(B = b) \text{MEC}(\mu_{X|b}, \mu_{Y|b}).$$

This yields the sharp universal bound

$$0 \leq \Gamma_B \leq \min\{H(X | B), H(Y | B)\}, \quad \chi_B \leq 2\kappa.$$

The constraint is nonredundant even for the simplest acyclic two-context cover. A support-minimal binary-overlap example has $\Gamma_B = 1/2$ bit. More strongly, for every integer $m \geq 2$ we construct an acyclic two-context instance with

$$\Gamma_B = \frac{1}{2} \log_2 m,$$

so the additive premium is unbounded as the local alphabets grow, while $\chi_B/\kappa \rightarrow 2$, showing asymptotic sharpness of the multiplicative bound.

The broad ingredients are not new: minimum-entropy coupling, support-constrained coupling problems, and local-versus-global consistency all have substantial prior literature. The narrow point of this note is to compare unconstrained minimum-entropy coupling with exact semantic gluing on shared coordinates and to quantify the resulting finite information premium even when global consistency is fully feasible and the cover is acyclic.

1 Introduction

A recurring local-to-global pattern is the following. Two local descriptions are individually correct, they agree statistically on what they share, and a global description exists. Nevertheless, a joint random source that generates the two local descriptions with minimal entropy need not respect their shared semantics sample by sample.

This distinction is invisible to ordinary minimum-entropy coupling (MEC). Given marginal laws μ on X and ν on Y , ordinary MEC minimizes

$$H(X, Y)$$

over all couplings of μ and ν [1, 2]. A coupling is free to pair any state of X with any state of Y provided the two marginals are correct.

Suppose instead that both local states contain a representation of the same overlap variable B . Write

$$r : X \rightarrow B, \quad s : Y \rightarrow B$$

for the two restriction maps. A genuine global realization must satisfy

$$r(X) = s(Y) \quad \text{almost surely.}$$

This is not an additional marginal condition. It is a support condition on the coupling itself.

The central question of this note is therefore:

How much additional Shannon entropy can be forced solely by requiring a common random coupling to possess one globally consistent interpretation on the overlap?

We compare two quantities. The first is ordinary MEC, denoted κ . The second, denoted χ_B , is MEC restricted to couplings that glue the overlap exactly. Their difference $\Gamma_B = \chi_B - \kappa$ measures the cost of global consistency relative to semantically unconstrained marginal generation.

The distinction is elementary to state, but it is not vacuous. We prove an exact conditional decomposition of χ_B , a sharp entropy bound on Γ_B , a support-minimal strict example, and an unbounded family on the acyclic cover

$$\{A, B\}, \quad \{B, C\}.$$

Thus acyclicity can guarantee that gluing is possible without making gluing free in the minimum-entropy sense.

This should be separated from two established bodies of work. First, MEC and its algorithmic approximations are well developed [1, 2, 3, 4]. Second, local-to-global consistency is classical in probability, databases, constraint systems, and contextuality. Acyclicity characterizes broad classes in which pairwise/local consistency guarantees global consistency [5], while contextuality can be formulated as an obstruction to global sections [6]. The present note does not claim either theme as new. It studies a different question inside the feasible regime: once a global extension exists, can exact gluing itself raise the least possible Shannon entropy of a common coupling?

2 Minimum-entropy coupling with an overlap constraint

Let X, Y, B be finite sets. Let $\mu \in \Delta(X)$ and $\nu \in \Delta(Y)$ be probability distributions, and let

$$r : X \rightarrow B, \quad s : Y \rightarrow B$$

be deterministic restriction maps.

The ordinary coupling polytope is

$$\mathcal{C}(\mu, \nu) := \{P \in \Delta(X \times Y) : P_X = \mu, P_Y = \nu\}.$$

Definition 1 (Unconstrained minimum-entropy coupling cost). *Define*

$$\kappa(\mu, \nu) := \min_{P \in \mathcal{C}(\mu, \nu)} H_P(X, Y).$$

Equivalently, $\kappa(\mu, \nu) = \text{MEC}(\mu, \nu)$.

The overlap-compatible coupling set is

$$\mathcal{C}_B(\mu, \nu) := \{P \in \mathcal{C}(\mu, \nu) : P(r(X) = s(Y)) = 1\}.$$

Definition 2 (Exact gluing cost). *Define*

$$\chi_B(\mu, \nu) := \begin{cases} \min_{P \in \mathcal{C}_B(\mu, \nu)} H_P(X, Y), & \mathcal{C}_B(\mu, \nu) \neq \emptyset, \\ +\infty, & \mathcal{C}_B(\mu, \nu) = \emptyset. \end{cases}$$

Definition 3 (Gluing entropy premium). *Whenever $\chi_B < \infty$, define*

$$\Gamma_B(\mu, \nu) := \chi_B(\mu, \nu) - \kappa(\mu, \nu) \geq 0.$$

The nonnegativity is immediate because $\mathcal{C}_B(\mu, \nu) \subseteq \mathcal{C}(\mu, \nu)$.

2.1 Feasibility of exact gluing

Let

$$\lambda_X := r_{\#}\mu, \quad \lambda_Y := s_{\#}\nu$$

be the two induced overlap laws.

Proposition 4 (Two-context gluing criterion). *The compatible coupling set $\mathcal{C}_B(\mu, \nu)$ is nonempty if and only if*

$$\lambda_X = \lambda_Y.$$

Proof. Necessity is immediate: if $r(X) = s(Y)$ almost surely, then the two random variables have the same law.

For sufficiency, write the common overlap law as λ . For every b with $\lambda(b) > 0$, let μ_b and ν_b denote the conditional laws of X and Y on the fibers

$$X_b := r^{-1}(b), \quad Y_b := s^{-1}(b).$$

Define a coupling by first sampling $B \sim \lambda$ and then, conditional on $B = b$, sampling $X \sim \mu_b$ and $Y \sim \nu_b$ independently. The resulting law has marginals μ and ν and is supported on pairs satisfying $r(x) = s(y)$. $\square$

Thus, for two contexts, overlap agreement removes the feasibility obstruction completely. The remaining question is quantitative rather than existential.

3 Exact fiberwise gluing formula

Assume from now on that

$$r_{\#}\mu = s_{\#}\nu = \lambda.$$

For $\lambda(b) > 0$, let μ_b and ν_b denote the normalized conditional laws on X_b and Y_b .

Theorem 5 (Exact fiberwise gluing formula). *For every finite two-context gluing problem,*

$$\boxed{\chi_B(\mu, \nu) = H(\lambda) + \sum_{b \in B} \lambda(b) \text{MEC}(\mu_b, \nu_b).} \quad (1)$$

Terms with $\lambda(b) = 0$ are understood to vanish.

Proof. Let $P \in \mathcal{C}_B(\mu, \nu)$. Since $r(X) = s(Y)$ almost surely, there is a well-defined common random variable

$$B = r(X) = s(Y).$$

By the chain rule,

$$H_P(X, Y) = H(B) + H_P(X, Y \mid B).$$

The overlap law is fixed and equals λ , hence

$$H_P(X, Y) = H(\lambda) + \sum_b \lambda(b) H_P(X, Y \mid B = b).$$

For each b , the conditional joint law is a coupling of μ_b and ν_b , so

$$H_P(X, Y \mid B = b) \geq \text{MEC}(\mu_b, \nu_b).$$

This proves the lower bound in (1).

Conversely, for each b choose an entropy-minimizing coupling P_b^* of μ_b and ν_b . Sample $B \sim \lambda$ and then sample $(X, Y) \sim P_b^*$ conditional on $B = b$. The resulting global law belongs to $\mathcal{C}_B(\mu, \nu)$ and attains the right-hand side of (1). $\square$

Remark 6. *Equation (1) shows that exact gluing is not a new coupling primitive inside each fiber. It decomposes into ordinary MEC problems after the common overlap value has been fixed. The distinction from unconstrained MEC lies in the prohibition against moving probability mass across different overlap fibers in order to lower the global joint entropy.*

4 Universal bounds

The exact formula immediately yields a sharp additive upper bound on the premium.

Theorem 7 (Conditional-entropy bound). *For every feasible two-context gluing problem,*

$$\boxed{0 \leq \Gamma_B \leq \min\{H(X \mid B), H(Y \mid B)\}.} \quad (2)$$

Consequently,

$$\boxed{\chi_B \leq 2\kappa.} \quad (3)$$

Proof. The lower bound follows from $\mathcal{C}_B \subseteq \mathcal{C}$.

For the upper bound, an independent coupling inside each overlap fiber gives

$$\text{MEC}(\mu_b, \nu_b) \leq H(\mu_b) + H(\nu_b).$$

Using (1),

$$\chi_B \leq H(B) + H(X \mid B) + H(Y \mid B).$$

Every ordinary coupling satisfies

$$H(X, Y) \geq \max\{H(X), H(Y)\},$$

so

$$\kappa \geq \max\{H(X), H(Y)\}.$$

Because B is a deterministic function of either local state,

$$H(X) = H(B) + H(X | B), \quad H(Y) = H(B) + H(Y | B).$$

Subtracting yields

$$\Gamma_B \leq H(X | B) + H(Y | B) - \max\{H(X | B), H(Y | B)\},$$

which is exactly (2).

Finally,

$$\min\{H(X | B), H(Y | B)\} \leq \min\{H(X), H(Y)\} \leq \kappa,$$

so $\chi_B = \kappa + \Gamma_B \leq 2\kappa$. □

Remark 8 (Mutual-information interpretation). *Since the two local marginals are fixed,*

$$H(X, Y) = H(X) + H(Y) - I(X; Y).$$

Hence

$$\Gamma_B = I_{\max}^{\text{free}}(X; Y) - I_{\max}^{\text{glued}}(X; Y),$$

where the first maximum ranges over all couplings and the second over overlap-compatible couplings. The premium is therefore exactly the maximal mutual information lost when one forbids semantically inconsistent pairings across overlap fibers.

4.1 When gluing is free

Proposition 9 (Elementary zero-premium criteria). *Each of the following conditions implies $\Gamma_B = 0$.*

- (i) $H(X | B) = 0$;
- (ii) $H(Y | B) = 0$;
- (iii) *there exists a deterministic map $f : X \rightarrow Y$ such that*

$$s(f(x)) = r(x) \quad \text{for all } x \in \text{supp } \mu, \quad f_{\#}\mu = \nu;$$

- (iv) *symmetrically, there exists an overlap-preserving deterministic map $g : Y \rightarrow X$ with $g_{\#}\nu = \mu$.*

Proof. Statements (i) and (ii) follow immediately from (2).

For (iii), couple $Y = f(X)$. The coupling is overlap-compatible and has

$$H(X, Y) = H(X).$$

Since Y is a deterministic function of X , $H(Y) \leq H(X)$, while every coupling obeys $H(X, Y) \geq H(X)$. Thus this compatible coupling is already a global MEC and $\chi_B = \kappa = H(X)$. Statement (iv) is symmetric. □

The exact characterization is taut but useful:

$$\Gamma_B = 0 \iff \text{at least one unconstrained MEC is overlap-compatible.}$$

The point of the strict examples below is that this need not happen even when gluing is perfectly feasible.

5 A support-minimal strict example

We now give the smallest binary-overlap witness in terms of nonzero local support sizes.

Let

$$B = \{0, 1\}.$$

Take three supported states on each side:

$$X = \{x_1, x_2, x_\star\}, \quad Y = \{y_\star, y_1, y_2\}.$$

Define the overlap maps by

$$\begin{aligned} r(x_1) = r(x_2) = 0, \quad r(x_\star) = 1, \\ s(y_\star) = 0, \quad s(y_1) = s(y_2) = 1. \end{aligned}$$

Assign local probabilities

$$\mu = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right), \quad \nu = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right),$$

in the orders displayed above. Both induce the fair overlap law

$$P(B = 0) = P(B = 1) = \frac{1}{2}.$$

Theorem 10 (Exact half-bit gluing premium). *For the instance above,*

$$\boxed{\kappa = \frac{3}{2}, \quad \chi_B = 2, \quad \Gamma_B = \frac{1}{2} \text{ bits.}} \tag{4}$$

Moreover, the upper bound in (2) is attained with equality.

Proof. The two local laws have the same multiset of atom masses,

$$\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right\}.$$

An unconstrained deterministic matching of equal-mass atoms therefore gives a coupling with entropy $3/2$ bits. No coupling can have entropy below either marginal entropy, and

$$H(\mu) = H(\nu) = \frac{3}{2}.$$

Hence $\kappa = 3/2$.

For exact gluing, conditional on $B = 0$, the X -fiber is uniform on two points while the Y -fiber is deterministic. Therefore

$$\text{MEC}(\mu_0, \nu_0) = 1.$$

Conditional on $B = 1$ the roles are reversed, so

$$\text{MEC}(\mu_1, \nu_1) = 1.$$

Equation (1) gives

$$\chi_B = 1 + \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 2.$$

Thus $\Gamma_B = 1/2$ bit.

Finally,

$$H(X | B) = H(Y | B) = \frac{1}{2},$$

so the bound in (2) is saturated. □

The compatible coupling is in fact unique: its four nonzero atoms have masses $1/4$ each,

$$(x_1, y_\star), (x_2, y_\star), (x_\star, y_1), (x_\star, y_2).$$

The unconstrained MEC instead pairs the unique $1/2$ atom on the X side with the unique $1/2$ atom on the Y side, even though these two states carry opposite overlap values. That cross-fiber pairing is precisely the entropy-saving move forbidden by exact gluing.

Proposition 11 (Support minimality for binary overlap). *Assume both overlap values have positive probability. If $\Gamma_B > 0$, then*

$$|\text{supp } \mu| \geq 3, \quad |\text{supp } \nu| \geq 3.$$

Hence the preceding 3×3 example is support-minimal.

Proof. If $|\text{supp } \mu| \leq 2$ while both overlap values have positive probability, then exactly one supported X -state lies above each overlap value. Thus X is a deterministic function of B and

$$H(X | B) = 0.$$

By (2), $\Gamma_B = 0$. The same argument applies to Y . □

6 An unbounded sharp family on an acyclic cover

The half-bit example is the first member of a family in which the premium grows without bound.

Fix $m \geq 2$. Let

$$X_m = \{x_\star, x_1, \dots, x_m\}, \quad Y_m = \{y_\star, y_1, \dots, y_m\}.$$

Define

$$\begin{aligned} r(x_i) &= 0 \quad (1 \leq i \leq m), & r(x_\star) &= 1, \\ s(y_\star) &= 0, & s(y_i) &= 1 \quad (1 \leq i \leq m). \end{aligned}$$

Let

$$\begin{aligned} \mu_m(x_\star) &= \frac{1}{2}, & \mu_m(x_i) &= \frac{1}{2m}, \\ \nu_m(y_\star) &= \frac{1}{2}, & \nu_m(y_i) &= \frac{1}{2m}. \end{aligned}$$

Again the overlap is fair.

Theorem 12 (Unbounded acyclic gluing premium). *For every integer $m \geq 2$,*

$$\boxed{\kappa_m = 1 + \frac{1}{2} \log_2 m,} \tag{5}$$

$$\boxed{\chi_{B,m} = 1 + \log_2 m,} \tag{6}$$

and therefore

$$\boxed{\Gamma_{B,m} = \frac{1}{2} \log_2 m.} \tag{7}$$

In particular, the additive gluing premium is unbounded as $m \rightarrow \infty$, and

$$\boxed{\frac{\chi_{B,m}}{\kappa_m} \rightarrow 2.} \tag{8}$$

Thus both the additive bound (2) and the multiplicative bound (3) are sharp in the natural asymptotic sense.

Proof. The two local distributions have the identical atom-mass multiset

$$\left\{ \frac{1}{2}, \underbrace{\frac{1}{2m}, \dots, \frac{1}{2m}}_{m \text{ times}} \right\}.$$

Therefore an unconstrained deterministic matching of equal-mass atoms attains the marginal entropy, which is the universal lower bound for a coupling. Hence

$$\kappa_m = H(\mu_m) = 1 + \frac{1}{2} \log_2 m.$$

For exact gluing, conditional on $B = 0$, X_m is uniform on m points and Y_m is deterministic, so the conditional MEC is $\log_2 m$. Conditional on $B = 1$, the roles reverse and the conditional MEC is again $\log_2 m$. Equation (1) gives

$$\chi_{B,m} = 1 + \frac{1}{2} \log_2 m + \frac{1}{2} \log_2 m = 1 + \log_2 m.$$

Subtracting yields (7), and (8) follows directly.

Moreover,

$$H(X_m | B) = H(Y_m | B) = \frac{1}{2} \log_2 m,$$

so (2) is attained with equality for every m . $\square$

This family makes the structural point especially clear. Nothing obstructs global consistency: the two local laws agree exactly on the overlap, and the cover has only two contexts. Yet allowing an unconstrained common source to reinterpret overlap labels across the two local decoders can save an arbitrarily large number of bits.

7 From two contexts to local-to-global consistency

The two-context construction may be embedded into the standard local-to-global language. Let the contexts be

$$C_1 = \{A, B\}, \quad C_2 = \{B, C\}.$$

A local state $x \in X$ is an assignment on C_1 and a local state $y \in Y$ is an assignment on C_2 . The maps r and s are restriction to the shared coordinate B .

A globally valid pair must satisfy

$$x|_B = y|_B.$$

The hypergraph with edges C_1, C_2 is acyclic. Classical local-to-global results imply that, in acyclic covers, appropriate local consistency conditions suffice for existence of a global object; in the probabilistic two-context case, equality of the overlap marginals is already necessary and sufficient [5]. By contrast, cyclic measurement covers may exhibit genuine obstructions to global sections, as emphasized in contextuality theory [6].

The premium Γ_B distinguishes a different phenomenon. It is defined only after feasibility has been secured. Thus one obtains three regimes:

$\Gamma_B = 0$: free gluing, $0 < \Gamma_B < \infty$: costly but feasible gluing, $\chi_B = +\infty$: impossible gluing.	(9)
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The second regime is the focus here. The family in Theorem 5 shows that it occurs already on the simplest acyclic cover and that its information cost need not be bounded by a universal constant.

8 General support-constrained formulation

The overlap problem is a special case of a more general support restriction.

Let $X_1, \dots, X_k$ be finite local state spaces with prescribed marginals $\mu_1, \dots, \mu_k$. Let

$$\mathbf{G} \subseteq X_1 \times \dots \times X_k$$

be the set of globally compatible tuples. Define

$$\kappa(\mu_1, \dots, \mu_k) := \min_{P: P_i = \mu_i} H(P),$$

and

$$\chi_{\mathbf{G}} := \min_{\substack{P: P_i = \mu_i \\ \text{supp } P \subseteq \mathbf{G}}} H(P),$$

with $\chi_{\mathbf{G}} = +\infty$ when the constrained set is empty. The general compatibility premium is

$$\Gamma_{\mathbf{G}} := \chi_{\mathbf{G}} - \kappa$$

whenever $\chi_{\mathbf{G}} < \infty$.

At this level the construction is simply minimum-entropy coupling over a smaller feasible polytope. No claim is made that this generic support-constrained optimization form is novel. The two-context overlap problem is useful because its compatibility relation is canonical, its constrained optimum admits the exact formula (1), and the resulting premium has sharp structural bounds and explicit extremal families.

A response-tree compatibility constraint for adaptive controlled processes provides another, more structured instance: the allowed support consists of tuples of intervention transcripts that can arise from one common deterministic response tree. In that setting the corresponding difference between ordinary transcript MEC and response-tree-compatible MEC measures a causal-consistency entropy premium [7]. The present two-context analysis isolates the static gluing mechanism without temporal structure.

9 Relation to existing work and claim boundary

The surrounding mathematical ingredients have substantial prior art.

Minimum-entropy coupling. The MEC problem, its hardness, structural properties, and approximation algorithms are well established [1, 2, 3]. Recent iterative and partition-based methods address large-support and autoregressive distributions [4]. We therefore do not present κ as a new primitive.

Local and global consistency. The distinction between locally consistent data and globally consistent realizations is classical. Vorob'ev-type results and their modern generalizations characterize acyclic schemes for which local consistency guarantees global consistency; Atserias and Kolaitis formulate a broad semiring version of this principle [5]. In a different language, Abramsky and Brandenburger characterize contextuality through obstructions to global sections [6]. We do not claim the local-to-global distinction itself as new.

What is narrower here. The specific comparison studied in this note is between

$$\min H(\text{arbitrary coupling of local-state laws})$$

and

$$\min H(\text{coupling with exact samplewise overlap agreement}).$$

The contribution is the exact fiberwise formula, the sharp conditional-entropy bound, the support-minimal strict witness, and the unbounded family showing a positive and arbitrarily large premium on a fully feasible acyclic two-context cover.

A literature search cannot establish absolute historical priority. The appropriate claim is therefore structural rather than historical: the results above identify and quantify a finite information cost of exact gluing that is distinct from the better-known question of whether a global extension exists at all.

10 Open problems

The elementary two-context theory leaves several natural directions.

- (1) **Exact zero-premium characterization.** Beyond the tautological condition that some unconstrained MEC is already compatible, characterize $\Gamma_B = 0$ directly from the fiber mass profiles.
- (2) **General acyclic covers.** For a junction tree of more than two contexts, determine whether the minimum entropy of a global extension admits a recursive MEC formula over separators, and obtain sharp analogues of (2).
- (3) **Cycles and feasibility versus cost.** Separate quantitatively the finite premium inside the globally consistent regime from genuine local-to-global obstructions on cyclic covers.
- (4) **Complexity.** Ordinary MEC is NP-hard. Determine the complexity of exact support-constrained MEC for fixed overlap structure and of deciding whether $\Gamma_B > 0$.
- (5) **Approximate gluing.** Replace the hard constraint $r(X) = s(Y)$ almost surely by a mismatch tolerance

$$P(r(X) \neq s(Y)) \leq \varepsilon$$

and study the resulting entropy-versus-consistency tradeoff.

- (6) **Multi-marginal semantic gluing.** For general compatibility relations $\mathbf{G}$, identify structural conditions under which $\Gamma_{\mathbf{G}}$ is additive, subadditive, bounded by separator entropies, or decomposable by dynamic programming.

11 Conclusion

Ordinary minimum-entropy coupling asks how efficiently several marginal laws can share randomness. Exact gluing asks for more: states that refer to the same overlap must agree on that overlap in every joint realization.

For two finite contexts this additional requirement is completely tractable at the level of principle. Once the overlap laws agree, a global coupling always exists, and its minimum entropy decomposes fiberwise as

$$\chi_B = H(B) + \sum_b P(b) \text{MEC}(\mu_{X|b}, \mu_{Y|b}).$$

Nevertheless, feasibility does not imply zero cost. The gluing premium obeys

$$0 \leq \Gamma_B \leq \min\{H(X \mid B), H(Y \mid B)\},$$

and this bound is sharp. On the acyclic cover $\{A, B\}, \{B, C\}$, the premium can equal $1/2$ bit in the smallest binary-overlap witness and can grow as $\frac{1}{2} \log_2 m$ in a simple family, while the constrained-to-unconstrained entropy ratio approaches two.

The resulting distinction is concise:

$$\text{local marginal compatibility} \neq \text{entropy-free global gluing.}$$

A global object may exist with no consistency obstruction and yet require strictly more shared Shannon information than is needed to reproduce its local descriptions separately.

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